

MATH3004 Industrial Project

Project Proposal

Benchmark Dataset Generator for Machine Learning-as-a-Service Ecosystems

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1 Background

The rapid growth of the Internet of Things (IoT) has greatly increased the demand for machine learning (ML) systems capable of processing large volumes of data. For Example, modern applications in domains such as healthcare, transportation, manufacturing, and smart cities rely on ML models to analyse sensor data, detect patterns, and support automated decision-making (Lee et al. 2015; IoT For All 2025). As these systems continue to grow, the integration of machine learning models has shifted towards cloud-based platforms.

Machine Learning-as-a-Service (MLaaS) is a tool that allows users to utilise machine learning models without maintaining their own infrastructure (Ribeiro et al. 2015). Major cloud providers such as Amazon SageMaker, Google Cloud AI, and Microsoft Azure Machine Learning offer services for tasks including image classification, text analysis, and forecasting (Noshiri et al. 2021). These services allow organisations to integrate machine learning capabilities cost-effectively.

As task complexity and data pipelines grow, a single, overarching ML model is often insufficient. To address this, multiple MLaaS services are combined to perform specialised tasks within a larger system, a concept known as **MLaaS composition** (Xie et al. 2024; Kanneganti et al. 2025) as shown in figure 1. For example, a monitoring system may integrate image recognition, natural language processing, and predictive analytics models to analyse respective data types. In these systems, effectively selecting and combining appropriate ML services is an important challenge. Different ML services may vary significantly in characteristics such as accuracy, training cost, computational requirements, and reliability.

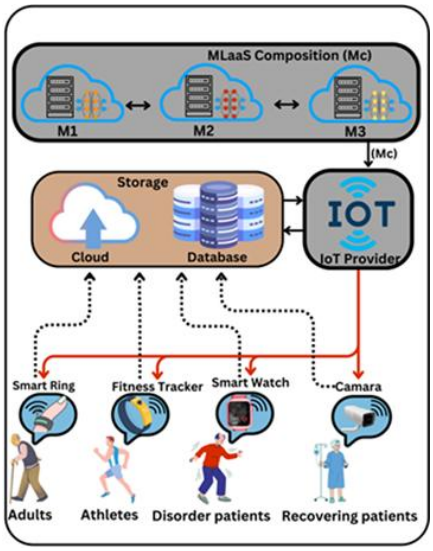


Figure 1: Illustration of MLaaS Composition

Recent ML studies have focused on this area, identifying the best combinations of ML services that optimise performance while meeting user-specified constraints such as latency and resource usage (Kanneganti et al. 2025). However, a significant challenge in this field is the scarcity of publicly available datasets describing the behaviour of ML services under different conditions. Many MLaaS providers reveal only limited information about their internal models and performance metrics due to commercial and proprietary restrictions (Noshiri et al. 2021). This restricts research to expensive or small-scale experiments.

A good contrast is the field of web services. This area has benefited from several publicly available benchmark datasets, which have enabled the development and evaluation of numerous algorithms for service discovery, ranking, and composition. The QWS dataset provides multiple Quality-of-Service (QoS) attributes for thousands of services and has become a widely used benchmark for service selection research (Al-Masri and Mahmoud 2008). Similarly, the WS-DREAM project released large-scale QoS datasets consisting of millions of service invocation records, supporting research on service recommendation and QoS prediction (Zheng et al. 2013) However, web-service datasets do not capture the unique characteristics of machine learning services. A ML version of these datasets would benefit the AI field in the same way.

To address this gap, this project proposes developing a benchmark dataset generator for Machine Learning-as-a-Service ecosystems. The framework will gather and simulate ML service data by training and testing models across a range of sources and configurations. For each service, a wide range of causal attributes and quality-of-service metrics will be recorded. These datasets will provide a detailed view of ML services to study service selection and composition in MLaaS environments.

2 Objectives of the research

The objective of this project is to design and implement a framework capable of generating datasets that represent heterogeneous Machine Learning-as-a-Service providers.

The project aims to achieve the following objectives:

1. Design a configurable dataset generation framework that supports a diverse range of machine learning task families machine learning services with diverse model configurations and training conditions.
2. Integrate both model training and inference workflows, enabling the framework to collect realistic performance measurements from commonly used machine learning models.
3. Generate datasets that capture both functional attributes and quality-of-service metrics describing the behaviour of individual ML service's and ML service compositions.
4. Evaluate and comparing existing selection/compositions algorithms on the produced datasets and comparing results to previous research.
5. Provide documentation and tools that allow researchers to reproduce experiments and extend the generated datasets.

3 Significance of the research

As ML services become more common, selecting the right ones to integrate is becoming more important. Despite this, research in ML service selection and composition is limited by the lack of available datasets describing ML service behaviour (Noshiri et al. 2021; Kanneganti et al. 2025). ML services are highly dependent on their configurations, such as training data, model architecture, and hyperparameters. These factors influence a service's performance and must be considered when evaluating combinations.

The proposed dataset generator framework will provide a reproducible method for generating datasets that describe machine learning services. Utilising a wide range of configurations, the framework will capture how service performance differs across different tasks and conditions. This will allow researchers to study relationships between model configurations, resource usage, and service performance.

The generated datasets will also support research on **machine learning service composition**, where multiple services are integrated to perform complex tasks. For example, a system may combine services for data preprocessing, classification, and summarisation. Understanding how different services interact within a composition is important for designing reliable AI pipelines.

The framework will also provide a place to evaluate algorithms for service selection to optimise QoS attributes for a given task. Similar benchmark datasets have played an important role in advancing research on web service selection and recommendation systems (Al-Masri and Mahmoud 2008; Zheng et al. 2013). Developing a similar benchmark resource for ML services could significantly accelerate research in this area.

4 Research methods

To generate a diverse MLaaS dataset, we follow several stages: task selection, service simulation, model training, and dataset construction.

4.1 Framework Design

A modular Python framework will be used to generate the datasets. This will allow researchers to configure datasets, models, and training parameters to evaluate a wide variety of service behaviours.

Each ML service instance will represent a provider offering a specific machine learning capability. Services will be instantiated using commonly used machine learning models and datasets and may involve either inference using pretrained models or training through fine-tuning or standard training procedures.

During execution, the framework will record measurements describing the behaviour of each service, including functional model attributes and quality-of-service metrics observed during training or inference.

4.2 Task and Dataset Selection

To represent the diversity of real-world ML services, the framework will support multiple tasks, as seen in table 1. As highlighted, these tasks will be executed using both pretrained models for inference and models trained or fine-tuned during experimentation. This allows for capture of realistic service behaviours observed in modern MLaaS environments.

Table 1: Task families supported by the MLaaS benchmark generator.

Task family	Task family
Image Classification	Zero-shot Text Classification
Text Classification	Sentence Similarity / Pair Scoring
Tabular Regression	Fill Mask (Masked Language Modelling)
Clustering (Tabular)	Summarisation
Text Generation	Image Generation (Text-to-Image)
Image-to-Text	Text-to-Speech
Token Classification	Question Answering

4.3 Service Simulation

Each ML service instance will represent a provider offering a specific machine learning capability. Services will be instantiated using machine learning models executed under different configurations, allowing the framework to evaluate a wide range of service behaviours. Services will differ across dimensions like:

- model architecture
- training hyperparameters
- dataset partitions and data distributions
- computational resource constraints

Examples of these parameters include learning rate, batch size, number of training epochs, optimisation algorithms, and model-specific settings. By varying these parameters, the framework will generate heterogeneous service instances that reflect differences in capability, performance, and computational requirements.

4.4 Model Training and Execution

The dataset will record machine learning services using both **training-based and inference-based workflows**. Training-based services will involve training or fine-tuning models using selected datasets and parameter configurations. Inference-based services will utilise pretrained models to generate outputs.

Classical machine learning algorithms and pretrained transformer models will be used to represent ML services performing a wide range of tasks.

4.5 Quality-of-Service Measurement

For each ML service, the framework will record quality-of-service metrics such as:

- model performance metrics (e.g., accuracy, F1 score, loss)
- training time and computational cost
- inference latency
- resource usage such as CPU or GPU memory consumption
- reliability indicators such as training completion status or failure events

These metrics provide a detailed description of the service's behaviour and allow comparisons between different service configurations.

4.6 Dataset and Composition Generation

The collected information will be stored in a normalised SQL database describing the behaviour of each service. Each service will record all previously set parameters and attributes, such as task type, dataset used, model architecture, configuration parameters, and quality-of-service metrics. These records will form a benchmark dataset representing ML services.

In addition to individual service records, the framework will also generate a dataset representing **service compositions**, where multiple ML services are combined to perform a larger task. The framework will record the interactions between services and the resulting system-level performance, enabling analysis of how different service combinations affect overall performance. These datasets will support future research on ML service selection, orchestration, and optimisation within MLaaS environments.

5 Expected outcomes

The expected outcomes of this project include:

- a configurable framework for generating benchmark datasets representing ML services
- a dataset describing machine learning services across multiple task domains
- a dataset describing compositions of ML services
- documentation describing the dataset schema and framework architecture
- tools for generating additional datasets using different configurations

The project may also contribute a research publication describing the framework and datasets.

The resulting framework and datasets will provide a foundation for future research on machine learning service selection, service composition, and performance optimisation in distributed AI systems.

6 References

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